

# FIXNOW

## Software Requirements Specification

### AI-Powered Home Service Booking Platform

| Item             | Value                    |
|------------------|--------------------------|
| Project Name     | FixNow                   |
| Document Version | 1.0                      |
| Document Status  | Approved for Iteration 1 |
| Backend          | Java 17/21 + Spring Boot |
| Database         | MySQL                    |
| Frontend         | React.js                 |
| Future Clients   | Android / iOS            |

Master requirements baseline for backend-first development and future UI integration.

# 1. Introduction

## 1.1 Purpose

FixNow is an AI-powered home-service platform designed to connect customers with verified home-service professionals. The platform allows customers to select a required service, describe their problem, upload photos or videos, obtain AI-assisted diagnosis and cost estimation, discover suitable workers, create bookings, track the worker during an active service, make payments, and provide ratings and reviews.

Workers can register, complete verification, manage professional information, select services, manage availability, receive service requests, accept or reject jobs, update job status, share location during active bookings, and view work and earnings.

The system will use a UI-independent Spring Boot REST backend so the same backend can support React Web and future mobile applications.

## 2. Product Vision

FixNow aims to make home-service booking faster, safer, more transparent, easier to manage, and more trustworthy.

Verified Workers + Service Marketplace + AI-Assisted Diagnosis + Cost Estimation + Booking + Tracking + Online Payment + Reviews

## 3. Problem Statement

### 3.1 Customer Problems

- Finding trustworthy service professionals.
- Verifying worker credentials.
- Finding workers near their location.
- Understanding likely repair/service cost.
- Explaining technical problems.
- Comparing available workers.
- Scheduling services.
- Knowing when the worker will arrive.
- Making secure payments.
- Tracking completed services and payments.
- Evaluating workers using previous customer experiences.

### 3.2 Worker Problems

- Finding customers.
- Getting consistent service requests.
- Establishing an online professional identity.
- Demonstrating skills and certifications.
- Managing service availability.
- Managing incoming jobs.
- Building customer trust.
- Tracking completed work and earnings.

## 4. Objectives

- Connect customers with suitable home-service workers.
- Provide verified worker profiles.

- Allow customers to select required services.
- Allow customers to describe service problems.
- Allow image/video uploads.
- Provide AI-assisted problem analysis and cost estimation.
- Recommend suitable services/workers.
- Provide structured booking.
- Allow workers to accept/reject jobs.
- Track booking status and active worker location.
- Support online payment.
- Generate payment/invoice records.
- Allow ratings and reviews.
- Provide worker profiles and job management.
- Maintain a secure, scalable backend.
- Keep backend independent of frontend implementation.

## 5. Scope

### 5.1 In Scope — Iteration 1

| Area           | Scope                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication | Registration, login, JWT authentication, role-based authorization, password security.                                                       |
| Customer       | Profile, addresses, service discovery, requests, problem description, media upload, AI diagnosis, cost estimation, worker discovery.        |
| Worker         | Registration, profile, KYC/documents, verification, services/skills, availability, requests, accept/reject, job status, location, earnings. |
| Admin          | Authentication, worker verification, basic user/service management, booking/payment monitoring.                                             |

### 5.2 Out of Scope for Iteration 1

- Real-time WebSocket chat.
- Advanced notification infrastructure.
- Advanced complaint management.
- Advanced analytics.
- Automated worker payouts/settlements.
- Wallet and subscription systems.
- Coupon system.
- Advanced AI history/learning.
- Advanced recommendation engine.
- Native Android application.
- Native iOS application.

## 6. Stakeholders

| Stakeholder           | Responsibility                   |
|-----------------------|----------------------------------|
| Customer              | Requests and books home services |
| Worker                | Provides home services           |
| Admin                 | Manages platform operations      |
| AI Service            | Problem analysis and estimation  |
| Payment Gateway       | Processes payments               |
| Maps/Location Service | Location functionality           |
| File Storage          | Stores uploaded files/documents  |
| Email/SMS Provider    | Communication/OTP where required |

## 7. User Roles

**Customer:** Finds and books services.

**Worker:** Provides services and manages jobs.

**Admin:** Manages platform operations and verification.

## 8. Customer Functional Requirements

| ID     | Requirement       | Description                                                                                                    |
|--------|-------------------|----------------------------------------------------------------------------------------------------------------|
| FR-C01 | Registration      | Allow customers to create accounts using required identity/authentication information.                         |
| FR-C02 | Login             | Allow customers to authenticate securely.                                                                      |
| FR-C03 | Profile           | Allow customers to view and update profiles.                                                                   |
| FR-C04 | Addresses         | Add, edit, delete and select a default address.                                                                |
| FR-C05 | Service Discovery | Browse configurable services such as Electrician, Plumber, AC Repair, Carpenter, Painter and Appliance Repair. |
| FR-C06 | Service Request   | Create a request with required service and problem description.                                                |
| FR-C07 | Media Upload      | Attach images/videos to a service request.                                                                     |
| FR-C08 | AI Diagnosis      | Submit problem information/media for AI-assisted analysis.                                                     |
| FR-C09 | AI Estimate       | Return problem type, cost range, duration, confidence, recommended service and, where supported, urgency.      |
| FR-C10 | Worker Discovery  | Return suitable workers using service, skills, experience, distance, availability, rating and verification.    |
| FR-C11 | Worker Profile    | View professional, experience, services, verification and rating information.                                  |
| FR-C12 | Booking           | Select a worker and create a booking.                                                                          |
| FR-C13 | Booking Status    | View current booking status.                                                                                   |
| FR-C14 | Tracking          | View worker location during an eligible active booking.                                                        |
| FR-C15 | Payment           | Make payment using supported methods.                                                                          |
| FR-C16 | Invoice           | Access payment/invoice information after successful payment.                                                   |
| FR-C17 | Review            | Submit rating, written review and optional review tags.                                                        |

## 9. Worker Functional Requirements

| ID     | Requirement        | Description                                                                                 |
|--------|--------------------|---------------------------------------------------------------------------------------------|
| FR-W01 | Registration       | Workers can register as service professionals.                                              |
| FR-W02 | Profile            | Workers maintain profile, experience, skills/services, rating and verification information. |
| FR-W03 | KYC                | Workers submit required identity and professional documents.                                |
| FR-W04 | Verification       | Support PENDING, APPROVED and REJECTED.                                                     |
| FR-W05 | Service Management | Associate workers with services they provide.                                               |
| FR-W06 | Service Pricing    | Maintain service-specific price.                                                            |
| FR-W07 | Availability       | Manage working availability.                                                                |
| FR-W08 | Service Requests   | View relevant service requests.                                                             |
| FR-W09 | Accept/Reject      | Accept or reject a job.                                                                     |
| FR-W10 | Job Status         | Update booking through valid states.                                                        |
| FR-W11 | Location Sharing   | Provide location during active tracked bookings.                                            |
| FR-W12 | Earnings           | View relevant earnings based on completed jobs/payments.                                    |

## 10. Admin Functional Requirements

| ID     | Requirement         | Description                                |
|--------|---------------------|--------------------------------------------|
| FR-A01 | Authentication      | Protected administrative access.           |
| FR-A02 | Worker Verification | View documents and approve/reject workers. |
| FR-A03 | User Management     | Manage platform users.                     |
| FR-A04 | Service Management  | Manage available services.                 |
| FR-A05 | Booking Monitoring  | View and monitor booking information.      |
| FR-A06 | Payment Monitoring  | View payment and transaction information.  |

## 11. Service Request Workflow

Select Service → Describe Problem → Upload Image/Video → AI Analysis → Diagnosis Result → Cost Estimate → Recommended Workers

## 12. Booking Workflow

Customer → Service Request → Worker Selection → Booking Creation → PENDING → Worker Accepts → ACCEPTED → ON\_THE\_WAY → STARTED → COMPLETED → PAYMENT → REVIEW

## 13. Booking Status Requirements

Supported initial statuses:

PENDING | ACCEPTED | ON\_THE\_WAY | STARTED | COMPLETED | CANCELLED

- Validate status transitions.

- Prevent invalid transitions.
- Maintain status history.
- Record actor where required: CUSTOMER, WORKER or SYSTEM.

## 14. AI Module Requirements

The AI module may use customer descriptions and uploaded images/videos where supported. Possible output: Problem Type, Description, Estimated Cost, Estimated Duration, Confidence Score, Recommended Service and Urgency.

**Business Rule:** AI output is an estimate and does not automatically represent the final payable amount.

## 15. Worker Recommendation Requirements

Worker recommendation/filtering may use:

Service Match + Skills + Experience + Distance + Availability + Rating + Verification

The recommendation algorithm may evolve independently from the API contract.

## 16. Payment Requirements

- Payment creation.
- Payment status.
- Transaction reference.
- Payment method.
- Payment verification.
- Successful and failed payment handling.
- Refund status where applicable.

Supported methods may include Razorpay, UPI, Card, COD and Wallet, subject to the configured provider.

## 17. Review Requirements

- A customer can review an eligible completed booking.
- Review contains booking, customer, worker, rating, comment, optional tags and timestamps.
- Rating is between 1 and 5.
- One booking should not receive multiple customer reviews unless rules change.

## 18. Tracking Requirements

Booking → Tracking Session → Tracking Points

- Tracking points contain location and timestamp.
- Backend controls access to tracking information.
- Tracking should normally be active only during the relevant booking lifecycle.

## 19. File and Document Requirements

- Customer: service-request images and videos.
- Worker: identity documents and professional certificates.
- System: invoice/receipt documents.

Large files should preferably be stored outside MySQL, while the database stores references and metadata.

## 20. Data Model Requirements

1. users
2. customer\_profiles
3. worker\_profiles
4. services
5. worker\_services
6. addresses
7. service\_requests
8. attachments
9. diagnoses
10. worker\_availabilities
11. worker\_documents
12. bookings
13. booking\_status\_history
14. payments
15. transactions
16. invoices
17. tracking
18. tracking\_points
19. reviews

## 21. Core Relationships

USER  
■■■■ CUSTOMER\_PROFILE  
■ ■■■ ADDRESS  
■ ■■■ SERVICE\_REQUEST  
■ ■■■ SERVICE  
■ ■■■ ATTACHMENT  
■ ■■■ DIAGNOSIS  
■ ■■■ BOOKING  
■ ■■■ WORKER  
■ ■■■ STATUS\_HISTORY  
■ ■■■ PAYMENT → TRANSACTION → INVOICE  
■ ■■■ TRACKING  
■ ■■■ REVIEW  
■■■■ WORKER\_PROFILE  
■■■■ SERVICE  
■■■■ AVAILABILITY  
■■■■ DOCUMENT  
■■■■ LOCATION

Worker and Service form a many-to-many relationship through WORKER\_SERVICE.

## 22. Security Requirements

- JWT authentication.
- Secure password hashing.
- Role-based authorization.
- Protected endpoints.
- Request and ownership validation.
- Secure payment verification.
- HTTPS in deployed environments.
- Protection of sensitive KYC documents and service attachments.

## 23. Backend Independence

The backend shall not depend on React components, routes, UI pages, CSS, frontend state or frontend-specific data structures.

React Web / Android / iOS → REST API / JSON → Spring Boot Backend → MySQL



## 24. API Requirements

REST APIs shall use versioning such as:

```
/api/v1/
```

Example resource groups:

```
/api/v1/auth  
/api/v1/users  
/api/v1/customers  
/api/v1/workers  
/api/v1/services  
/api/v1/service-requests  
/api/v1/bookings  
/api/v1/payments  
/api/v1/reviews
```

Detailed API definitions will be documented separately after the database schema is finalized.

## 25. External Integrations

| Integration          | Purpose                                                     |
|----------------------|-------------------------------------------------------------|
| Gemini AI API        | AI-assisted diagnosis and estimation                        |
| Google Maps Platform | Location, distance, nearby workers, maps and tracking       |
| Razorpay             | Payment processing, verification and transaction references |
| SMTP / SMS           | Communication and OTP where required                        |

## 26. Non-Functional Requirements

### 26.1 Performance

- Login target: approximately 2–3 seconds under normal load.
- Service/worker search target: approximately 3 seconds under normal conditions.
- Efficient database queries.
- Pagination for large collections.
- Validate targets through testing.

### 26.2 Scalability

- Support increasing users, workers, bookings and API traffic.
- Support future mobile applications.
- Allow additional services and integrations.

### 26.3 Maintainability

```
Controller → Service → Repository → Database
```

- Separate DTOs, validation, exception handling, security and external integrations appropriately.

### 26.4 Reliability

- Validate requests.
- Handle failures gracefully.
- Prevent inconsistent booking states.
- Prevent incorrect payment states.
- Maintain booking history and transaction records.

## 27. Business Rules

| ID    | Rule                | Requirement                                                                                |
|-------|---------------------|--------------------------------------------------------------------------------------------|
| BR-01 | Unique Account      | Email and phone shall be unique according to the authentication design.                    |
| BR-02 | Worker Verification | Worker verification status determines eligibility for normal platform operations.          |
| BR-03 | Service Association | A worker may provide multiple services and a service may have multiple workers.            |
| BR-04 | Booking Ownership   | A booking must reference valid customer, worker and service-request information.           |
| BR-05 | Status Validation   | A booking cannot move directly to an invalid lifecycle state.                              |
| BR-06 | Payment             | A booking must not be marked successfully paid without valid payment confirmation.         |
| BR-07 | Review              | A customer may review an eligible completed booking.                                       |
| BR-08 | AI Estimate         | AI cost is an estimate and does not automatically determine the final payable amount.      |
| BR-09 | Tracking            | Tracking information is only accessible to authorized participants.                        |
| BR-10 | File Security       | KYC documents and service attachments shall not be publicly exposed without authorization. |

## 28. Assumptions

- Users have internet access.
- Workers provide valid information and documents.
- Customers provide valid service locations.
- Customers may provide images/videos for AI analysis.
- Workers may provide GPS location during active tracking.
- External AI, payment and maps services are available.
- File storage is available.
- Third-party API limitations may apply.

## 29. Constraints

- Internet connectivity.
- AI API availability.
- Payment gateway availability.
- Maps API availability.
- GPS availability.
- File storage availability.
- Third-party API quotas.
- Email/SMS availability where used.

## 30. Technology Stack

| Area                  | Technology                                                                            |
|-----------------------|---------------------------------------------------------------------------------------|
| Frontend              | React.js, JavaScript, HTML5, CSS3, Tailwind CSS / Bootstrap                           |
| Backend               | Java 17/21, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Security, JWT |
| Database              | MySQL                                                                                 |
| Build                 | Maven or Gradle                                                                       |
| Development & Testing | IntelliJ IDEA, Postman, Git, GitHub                                                   |

## 31. Iteration 1 Implementation Scope

Project Setup → Database Configuration → Authentication → User/Customer/Worker Profiles → Service Management → Worker Services → Worker KYC → Worker Availability → Customer Addresses → Service Requests → Attachments → AI Diagnosis → Worker Discovery → Booking → Booking Status History → Basic Tracking → Payment → Transaction → Invoice → Review → Testing

## 32. Future Iterations

**Iteration 2:** Real-time WebSocket Chat, Notifications, Advanced GPS Tracking, Complaints and Advanced Admin Management.

**Future:** Worker Payouts, Advanced AI, Advanced Recommendation, Coupons, Wallet, Subscription, Android App and iOS App.

## 33. Documentation & Development Artifacts

```
FixNow/  
■■■■ docs/  
■ ■■■ 01-SRS.md  
■ ■■■ 02-ERD.png  
■ ■■■ 03-Database-Schema.md  
■ ■■■ 04-API-Contract.md  
■ ■■■ 05-LLD.md  
■ ■■■ 06-Implementation-Plan.md  
■■■■ backend/  
■■■■ frontend/  
■■■■ README.md  
■■■■ .gitignore
```

## 34. Development Lifecycle

SRS → Relationship Design → ERD → Database Schema → API Contract → LLD → Spring Boot Setup → Backend Implementation → API Testing → React Frontend → Frontend Integration → System Testing → Deployment

## 35. Acceptance Criteria for Iteration 1

### Customer Flow

Registration → Login → Select Service → Service Request → Upload Media → AI Diagnosis/Estimate → Find Worker → Select Worker → Booking → Worker Accepts → On The Way → Service Starts → Service Completes → Payment → Invoice/Receipt → Review

### Worker Flow

Registration → KYC → Admin Verification → Select Services → Set Availability → Receive Request → Accept/Reject → Update Job Status → Share Location → Complete Job → View Earnings

## 36. Final Product Definition

**FixNow** is an AI-powered home-service booking platform that connects customers with verified service professionals and provides service discovery, AI-assisted diagnosis and estimation, worker recommendation, booking, status tracking, payment, invoicing and reviews through a secure, scalable and UI-independent Spring Boot backend.

## 37. Document Status

| Artifact               | Status    |
|------------------------|-----------|
| SRS                    | Completed |
| Relationship Design    | Completed |
| ERD                    | Completed |
| Database Schema        | Next      |
| API Contract           | Pending   |
| LLD                    | Pending   |
| Backend Implementation | Pending   |
| Testing                | Pending   |
| React Integration      | Pending   |